Environment material management process

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Revised contents	Drift	Review	Approval
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Revised Items	AS IS	TO BE

Agreement	QM	R&D					

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	Opinion:

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[Revision history]

Revision history summary			RS-EP-01	
			Revision	001
			Confidential	G1
Rev. no	**Rev. date**	**Simple description of changed contents**	**Author**	**Post**
001	2022.10.26	Newly established.	Mark.Kim	QM

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1. Purpose and scope

This process defines the items and procedures to be managed in order to comply with international hazardous substances environment standards for product developed and mass produced by our company.

2. Apply scope

Raw material & packing material that is used to product's manufacturing.

3. Definition of terms

3.1 hazardous substances

Substance that bad affect the global environment among substances contained or used in products and packing materials.

3.2 Controlled material

Substances that are prohibited from being contained or used in products and packing material according to domestic and global laws and regulations among hazardous substances.

3.3 Raw material

All of materials to composing products & packing.

3.4 Prohibited substance content analysis result (Usually ICP analysis result)

As the quantitative and qualitative analysis data of the content of hazardous substances analyzed through the analysis method with certain validity, it is usually ICP analysis is a globally valid method.

4. Relevant form

N/A

5. Responsibility and authority

5.1 Subcontractors or suppliers

-. Subcontractors conduct preliminary environmental evaluation of raw-materials and packing materials applied to new product development. To conduct a preliminary review of the materials to ensure that environmentally hazardous substances and regulated substances are within the permitted range should be carried out

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-. The results of the evaluation of environmentally regulated substances for the relevant materials must be submitted to the company upon completion of development, and the results of the evaluation of environmentally regulated substances must be recognized during the internationally recognized certification period. The results of experiments conducted by non-accredited organizations can be used as reference data, but cannot be used as data for certification. The results of experiments conducted by institutions can be used as reference data, but cannot be used as data for certification.

-. After the completion of development, the monitoring of environmentally hazardous substances in the mass production process must be carried out, and the environment of the raw and subsidiary materials. The evaluation must be submitted to the company under revision management on a one-year cycle.

5.2 Quality management division

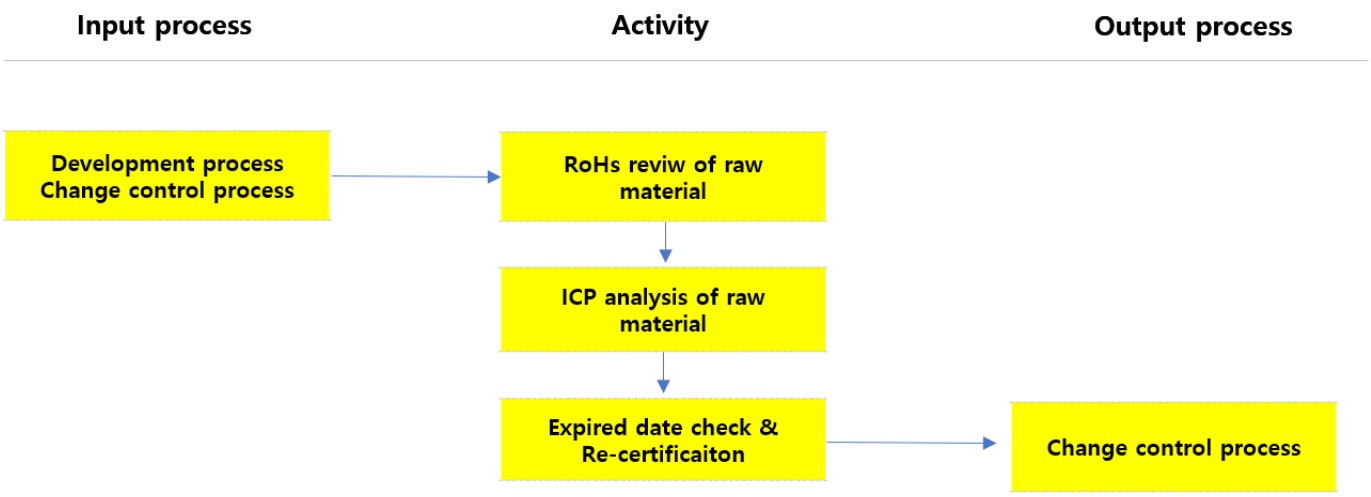
-. In the process of developing new products, our Quality Assurance Office has the duty and responsibility to check whether the raw and subsidiary materials developed by outsourced companies meet the international environmental regulatory substance management standards.

-. In the process of mass production, monitoring of environmentally controlled substances is secured from an outsourced company once/year for each raw and subsidiary material concerned.

-. If there is a change in the raw material, it is necessary to proceed with the qualification of whether the changed material contains harmful substances.

-. If necessary, request an external authorized agency to conduct analysis of environmentally hazardous substances (Cd, Hg, Pb, Cr+6, PBBS, PBDES) and outsource make sure to verify the partner data.

6. Process map



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7. Process description

In process	Input	Activity	Output	Out process	Freq.	Review	Approval
-. Development process -. Change control process	Product specification	In the development department, the BOM of the subcontractor When selecting, whether it contains environmentally harmful substances Prior review and compliance with international environmental regulations Check in advance whether there is a violation.	-. Select BOM -. Changed BOM	-	At event	Charge person	CTO CQO
-	-. Select BOM -. Changed BOM	-. The subcontractor conducts ICP analysis of the selected BOM. Environmentally hazardous substances within the standard Take a cognitive test and submit it to us. -. If the BOM needs to be changed, the subcontractor reports to the company in advance and conducts ICP analysis to guarantee the presence of hazardous substances.	-. PCN -. ICP Analysis result	-. Change control process	At event	Charge person	CQO
	ICP analysis report	During mass production, ICP for each BOM The report manages Expired data and Check if it is within the validity period After the expiration date, Request to renew your ICP report to keep it up to date should keep	-. Revised ICP Analysis report	-	At event	Chare person	CQO

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8. RoHS(Restriction of Hazardous Substance) Critiera

8.1 RoHS

Classification	Materials	CAS NO	Permission (PPM)	Criteria
RoHS	Cd	Cadmium	7440-43-9	5 ↓
	Pb	Lead	7439-92-1	100 ↓
	Hg	Mercury	7439-97-9	100 ↓
	Cr=6	Chromium 6+	18540-29-9	100 ↓
	PBBs	PBBs	13654-09-6	100 ↓
	PBDEs	PBDEs	1163-19-5	100 ↓

8.2 Restriction material

Classification	Material	CAS NO	PPM
Restriction material	Amides, amines	445409-27-8	1000
	Antimony and its compounds	7440-36-0	1000
	Arsenic and its compounds	7440-38-2	0
	Asbestos and its compounds	1332-21-4	0
	Azo dyes	495-48-7	30
	Beryllum and its compounds	7440-41-7	200
	Bismuth and its compounds	-	1000
	Chlorofluorocarbons	75-69-4	0
	Chloro parffins	63449-39-8	1000
	Ethlene Glycol Monoethyl Ether	110-80-5	5
	Ethylene Glycol Monomethyl Ether	109-86-4	5
	Formaldehydes	-	0.1
	Halons(Ozone Depleting Chemicals)	353-59-3	0
	Hydrochlorofluorocarbons (Ozone Depleting Chemicals)	75-45-6	0
	Hydrochlorofluoro carbons	-	1000
	Halogendated dioxins and furans	-	0
	Nikel and its compounds	7440-02-0	0
	Organo Tin and its compounds	7049-94-2	0
	Phthalates	-	1000
	Polychlorindatd biphenyls(PCBs)	1336-36-3	0
	Polychlorinated naphthanlenes(PCNs)	-	0
	Polychloroterphenyls(PCTs)	61788-33-8	0
	Polyvinyl choloride(PVC)	9002-86-2	0
	Selenium and its compounds	7782-49-2	1000
	Short chain chlorinated paraffins(SCCPs)	-	0

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8.2 Managing material

구분	물질		CAS-No.	ICP 분석 농도(ppm)
관리물질	Acronitrile	아크릴로 니트릴	107-13-1	-
	Acrylamide	아크릴 아미드	1979-06-01	-
	Barium and its compounds	바륨	7440-39-3	-
	Benzene	벤젠	81-43-4	-
	Carbon tetrachloride	사염화탄소	56-23-5	-
	Chloromethyl methyl ether	클로로메틸 메틸 에테르	107-30-2	-
	Chromium and its compounds	크롬	7758-97-6	-
	Cis-1,2-dichloroethtlene	시스-1,2-디크로로에틸	156-59-2	-
	Cobalt and its compounds	코발트	7440-48-4	-
	Copper and its compounds	구리	7440-50-8	-
	Cyanides	시안화합물	-	-
	Dichloromethane	디크로로메탄	75-09-002	-
	Dichloroethane	디크로로에탄	1330-21-6	-
	Dioxins	다이옥신	-	-
	Epichlorohydrin	에피클로로히드린	106-89-8	-
	Ethylene Glycol Monoethyl acetate	에틸렌그리콜 모노에틸 에테르 아세트산	111-15-9	-
	Ethylene Glycol Monomethyl acetate	에틸렌그리콜 모노메틸 에테르 아세트산	110-49-6	-
	Gold and its compounds (Au)	금	7440-75-5	-
	Hexachlorobenzene	헥사크로로벤젠	118-74-1	-
	Hydrofluorocarbons (Ozone Depleting Chemicals)	하이드로 프로카본(HFCs)	75-10-5	-
	Isocyanate	이소시아네이트	-	-

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	Acrylamide	아크릴 아미드	1979-06-01	-
	Barium and its compounds	바륨	7440-39-3	-
	Benzene	벤젠	81-43-4	-
	Carbon tetrachloride	사염화탄소	56-23-5	-
	Chloromethyl methyl ether	클로로메틸 메틸 에테르	107-30-2	-
	Chromium and its compounds	크롬	7758-97-6	-
	Cis-1,2-dichloroethtlene	시스-1,2-디크로로에틸	156-59-2	-
	Cobalt and its compounds	코발트	7440-48-4	-
	Copper and its compounds	구리	7440-50-8	-
	Cyanides	시안화합물	-	-
	Dichloromethane	디크로로메탄	75-09-002	-
	Dichloroethane	디크로로에탄	1330-21-6	-
	Dioxins	다이옥신	-	-
	Epichlorohydrin	에피클로로히드린	106-89-8	-
	Ethylene Glycol Monoethyl acetate	에틸렌그리콜 모노에틸 에테르 아세트산	111-15-9	-
	Ethylene Glycol Monomethyl acetate	에틸렌그리콜 모노메틸 에테르 아세트산	110-49-6	-
	Gold and its compounds (Au)	금	7440-75-5	-
	Hexachlorobenzene	헥사크로로벤젠	118-74-1	-
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	Isocyanate	이소시아네이트	-	-

구분	물질		CAS-No.	ICP 분석 농도(ppm)
관리물질	Magnesium	마그네슘	7439-95-4	-
	Manganese and its compounds	망간	7439-92-1	-
	Methyl bromide	브롬화메틸	74-83-9	-
	Methxylene		-	-
	Mirex(Perchlordecone)	미렉스	2385-85-5	-
	Palladium and its compounds (Pd)	팔라듐	7440-05-03	-
	Pentachlorophenol	펜타크로로페놀	87-86-5	-
	Perfluorocarbons (PFCs)	과불화 화합물	75-73-0	-
	Picric acid		88-89-1	-
	Radioactive substances	방사능 물질	-	-
	Silver and its compounds (Ag)	은	7440-212-4	-
	TBBP-A-bis Fr-720		-	-
	Tellurium	텔루르	13494-80-9	-
	Tetrachloroethylene	테크라크로로 에틸렌	127-18-4	-
	Thallium	탈륨	7440-28-0	-
	Trichloroethylene	트리크로로에틸렌	79-01-006	-
	1,1,1-trichloroethane	1,1,1-트리클로로에탄	56-23-6	-
	1,2-Dichloroethane	1,2-디크로로에탄	107-06-2	-
	1,2-Dichloroethylene	1,2-디크로로에틸렌	540-59-0	-
	1,2,2-Trichloroethane	1,2,2-트리크로로에탄	79-00-5	-
	1,3-Dichloropropene	1,3-디클로프로펜	542-75-6	-

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